(7) The *Contractor* shall provide all necessary accessories such as cable conduits, trunking and tray, power point (if required) and interconnection with the PCCW-HKT (or equivalent telecommunication company) outlets.

(8) The *Contractor* shall be responsible for all the E&M and associated works in the transformer room as required and requested by CLP for the permanent power supply shown in PS 42.03 and tender drawing. The E&M works shall include but not limited to supply, delivery to Site, installation and testing and commissioning of ventilation fan, air ductwork, MCB distribution board, switches, lightings, cable trunking and tray, earthing link, etc.

(9) The *Contractor* shall be responsible for all the minor civil requirements such as grouting-in and all civil finishing works as required by the public utilities. All holes required for drilled fixings shall be provided by the *Contractor*.

(10) The *Contractor* shall also be responsible for all deposits, service connection charges, fees, monthly/periodic rent/tariff/payment until the *works* are satisfactorily commissioned and taken over by the *Client*. The *Contractor* shall then settle all the outstanding charges with the relevant power company/PCCW and other related utilities prior to the change of consumership.

(11) The *Contractor* shall be responsible for the installation works as required by public utilities, services providers, etc, including temporary and permanent provisions such as fuel oils, power supply, etc.

Not Used 41.54 Not used.

Not Used 41.55 Not used.

Stormwater Pumps 41.56 (1) General

Comply (a) All the stormwater pumps shall be submersible pumps.

Comply (b) The *Contractor* shall perform the tests as specified under this Specification unless otherwise approved by the *Project Manager*.

Comply (c) All submersible pumps shall be suitable for pumping raw sewage containing rags, fibrous material and general waste matters including grit and other abrasives.

Comply (d) The pump supplied shall have stable and non-overloading characteristics.

Comply (e) Pumps shall be designed to meet the performance requirements and construction details stated in the Contract. In particular, the pumps shall be designed and constructed to ensure their hydraulic performance, robustness and reliability.

Comply (f) Pump performance shall be stable and free from cavitation, vibration and noise throughout the operating flow range and suction submergence.

Comply (g) Pumps shall be robust to withstand all forces associated with normal

and abnormal operating conditions including:

(i) normal starting and stopping against an opened or partially closed delivery valve;
(ii) reverse rotation due to backflow;
(iii) hydraulic transients due to power failure and emergency shutdown; and
(iv) impact and inertia forces associated with pumping sewage, particularly due to solids jamming elsewhere inside the pump.

Comply (h) Each component casting of the pump body shall have extra wall thickness to compensate for the corrosive and abrasive action of the sewage over 25 years operation in addition to the compliance with the strength requirement.

Comply (i) The Pump Suction Ability (Net Positive Suction Head Required-NPSHR) of a pump shall be compatible with that available to ensure efficient operation without cavitation over the full range of flows at all suction liquid levels and flow conditions. NPSHR shall be at least 2m less than the Net Positive Suction Head Actual, NPSHA, under any operating condition. The calculation shall be submitted to the *Project Manager* for acceptance.

Comply (j) The "Head-Quantity" characteristics of a pump shall be stable so that a tangent drawn of any point on the curve will be downwardly directed in the direction of increasing flow. "Head-Quantity" characteristics shall be designed to avoid large changes of flow with small changes of head. "Head-Quantity" characteristics shall also be capable of operating with the sump surcharged to its maximum level and pump motors shall be capable of meeting this condition.

Comply (k) Pumps shall be capable of running in reverse rotation at speeds of up to 125% of the maximum operating speed of the pump or the reverse speed developed due to any possibility of backflow, whichever is higher or otherwise accepted by the *Project Manager*.

Comply (l) Pumps in wet well shall be suitable for continuous running with very low levels of sewage in the sump whereas the pump motor and majority of the pump casing exposed above the sewage level.

Comply (m) Suction and discharge flanges shall be faced and drilled to BS EN 1092-2: PN16.

Comply (n) The minimum and maximum velocities at pump entrance and discharge shall follow pump supplier's recommended data for the prevention of solid settlement and excessive wear. In the absence of supplier's recommended data. Fluid velocity at pump entrance and discharge shall not exceed 3.6 m/s or otherwise accepted by the *Project Manager*.

(2) Head Loss and Pump Capacity

By other (a) The system losses are deemed to include the friction losses in the pipeworks. The *Contractor* shall estimate the station losses to be included in the total head of the pumps.

By other

Comply (b) Detailed calculations for the total head of the pumps shall be submitted for *Project Manager*'s acceptance. Supporting information including typical pump curves showing pumping heads, efficiencies and power absorbed at various flow rates of the pumps shall also be submitted.

Comply (c) The *Contractor* shall select rainwater pumps of appropriate size to suit all the flow requirements. The pump head requirements are indicative only and the *Contractor* shall design and submit the calculations for *Project Manager*'s acceptance.

Comply SWP-201 and SWP-202

Comply	Pump Capacity:	Each 1500 L/s
Comply	Head:	16m at 1500 L/s
Comply	Quantity:	2 nos. (1 Duty and 1 Standby)
Comply	Hydraulic Efficiency:	Not less than 82% at the duty point
Comply	Motor Efficiency:	Not less than 93% at the duty point
Suitable for VSD operation, inverter supplied by others	Speed Control:	Variable speed
	Starting Method:	Variable speed drive
Comply	Pump Starts Per Hour:	8 or more
Comply	Pump and Motor Rated Speed:	Less than 1000 RPM

SWP-801

Comply	Pump Capacity:	Each 400 L/s
Comply	Head:	3m at 400 L/s
Comply	Quantity:	1 no. (1 Duty)
Comply	Hydraulic Efficiency:	Not less than 73% at the duty point
Comply	Motor Efficiency:	Not less than 88% at the duty point
Comply	Speed Control:	Fixed speed
Comply	Starting Method:	Star-delta starter, DOL starter or Soft Starter
Comply	Pump Starts Per Hour:	8 or more
Comply	Pump and Motor Rated Speed:	Less than 1000 RPM

(3) Pump Construction

(a) Each pump shall be constructed with the following materials or better materials approved by the *Project Manager*: -

	Components	Materials
Comply	Pump casing:	Stainless Steel grade 316 to BS EN 10088, Cast Iron Grade EN-GLJ-250, Ductile Iron EN1563 EN-GJS-500-7 or accepted better material
Comply	Impeller:	Duplex Stainless Steel ASTM CD-4MCuN, Duplex Stainless Steel to AISI 329, Duplex Stainless steel Grade 1.4517 to BS EN 10213 or accepted better material
Comply	Shaft:	Stainless Steel grade 316 to BS EN 10088, Cast Stainless Steel to EN 1.4408, Stainless Steel

		ASTM-AISI 431, Stainless Steel grade 1.4021 or accepted better material
Comply	Seals:	Double mechanical shaft seals. For the mechanical seal on the pump side, the stationary faces shall be made of silicon carbide, tungsten carbide or accepted better material, and the rotating faces shall be made of tungsten carbide, silicon carbide or accepted better material. Moisture detector shall be provided to detect any intrusion of water.
Comply	Bolts and Nuts:	Stainless Steel, grade 316 to BS EN 10088, Stainless Steel CrNiMo A4 to BS EN 10088 or accepted better material
Comply	O rings:	Nitrite Rubber or accepted better material
Comply for lifting chain, Guide rails by other	Guide Rail and Lifting Chains:	Stainless Steel, grade 316 to BS EN 10088, Duplex Stainless steel Grade 1.4517 to BS EN 10213 or accepted better material

(b) Pump Casing

Comply (i) Casing shall be designed and constructed of such wall thickness and strength, not including the corrosion allowance, that the tensile stress in any part of the casing does not exceed 0.25 times the minimum ultimate tensile strength of the material when operating under the worst conditions of maximum speed and maximum suction head or otherwise accepted by the *Project Manager*. Casings shall be of construction to prevent undue vibration, to give long life abrasive conditions and to enable them to withstand shock loads caused by solids in dry weather flow.

Comply (ii) Casings shall be of a volute construction.

Comply (iii) Casings shall be provided with either integrally cast lifting lugs, drilled or cored lifting holes in external stiffening ribs, or collared eyebolts screwed into tapings in raised machined bosses. Lifting provisions shall be adequate for lifting the complete pump assembly.

(c) Impeller and Pump Shaft

Comply (i) Impeller blade shall be of non-clogging type and:

- Multiple vane, double shrouded type;
- Semi-open multi-van, back-swept leading edge;
- Screw centrifugal type;
- Free-flow centrifugal type;
- Closed multi-channel centrifugal type;
- Streamlined shape; or
- Proposed by the *Contractor* to the acceptance of the *Project Manager*

for transporting wastewater with solid and the internal passages shall be smoothly finished. Impellers shall be keyed and bolted to the shaft and the bolts positively locked to prevent loosening

or otherwise accepted by the *Project Manager*.

Bolted joints shall be provided with O-ring seals to prevent ingress of liquid into the impeller-shaft clearances and cavities.

Comply (ii) The impeller shall be capable of passing solid spheres up to 110mm diameter for SWP-201 and SWP-202, 105mm diameter for SWP-801, or 3 times of the bar screen bar spacing, whichever is lowest or otherwise accepted by the *Project Manager*.

Comply (iii) Shafts shall be designed to provide rigid support of the overhung impeller and to transmit loads without whip, vibration or undue deflection at all operating speeds under all operating conditions.

Comply (iv) Rotating elements of pumps shall be both statically and dynamically balanced before final assembly.

Comply (v) Impellers shall be keyed to the shafts so that they will not be loosened by reverse rotation and shall be readily removable.

Comply, double guide rails by other (vi) For the wet well installation, guide rail systems shall be completed with self-seating, discharge connector double guide rails, guide holders, lifting chain, chain bucket and lifting shackle and all accessories necessary for lifting the pump up to floor level.

Comply (vii) For the wet well installation, lifting chains shall include shackles for connection to pump lifting points. Lifting chains shall be at least 1.5m longer than the guide rails. Additional stainless steel rings shall be attached to the chain in an interval of 2m to facilitate the lifting of equipment by lifting appliances.

(d) Pump Bearings

Comply (i) Pump bearings shall be adequately rated, ball or roller type, oil or grease lubricated and shall be arranged to take all radial and axial loads during start-up and running of the pump.

Comply (ii) Pump bearings shall be greased lubricated and packed for one overhaul cycle of minimum 30,000 hours. The grease shall be of high quality type using barium complex as thickener. It shall have a maximum operating temperature of 150 °C and a dropping point of above 220 °C to DIN ISO 2176 or otherwise accepted by the *Project Manager*.

(e) Mechanical Seals

Comply (i) The mechanical seal shall be the self-adjusting non-water cooled type. It shall be automatic aligning type with independent self-aligning rotating and stationary parts. The wear surfaces of the seal shall be removable from the seal for inspection or repair easily without any special tools.

Comply

(ii) All the metal parts of seal shall be of stainless steel Grade 316 or otherwise accepted by the *Project Manager*. Shaft sealing shall be by independently mounted, mechanical seals shall have a solid tungsten carbide or silicon carbide rotating face running against a solid silicon or tungsten carbide stationary face. Any alternate materials for the rotating and stationary faces shall be proposed by the *Contractor* to the acceptance by the *Project Manager*. Seals with both faces of similar materials, or seals with bonded, soldered, or converted face surfaces are not equal or acceptable. The outer seal shall be a bellow-type construction, with a rubber bellow vulcanized to the rotating face and firmly clamped to the shaft, to keep contaminants from lodging in the stainless steel seal spring.

Comply

(iii) The outer mechanical seal shall be isolated from contaminants in the pumped media by a labyrinth-fit between the impeller backside and the volute backplate, as well as by pump-out grooves machined into the impeller backside and into the volute back plate, to prevent debris from reaching the shaft seal or otherwise accepted by the *Project Manager*.

Comply with sensor provided

(iv) Moisture sensor shall be provided for the oil chamber of the mechanical seal. The moisture sensor shall detect the failure of the primary seal and send an alarm signal to the SCADA system. All necessary transducer, interface, controllers and accessories shall also be included.

By other

(f) Pump Motor

Comply

(i) In addition to the requirements stipulated in Clause 3.3.1 and 3.3.2 of the GSEM, the pump motor provided shall also comply with the requirements specified in this Particular Specification.

Comply

(ii) The motor shall be designed for driving submersible sewage pumps. The motor casing shall be of cast iron grade 220 EN-GJL-250 with protective epoxy coating, cast Iron Gr250 with coating or accepted better material.

Comply

(iii) Motors shall have IP68 protection up to 20m submergence unless otherwise accepted by the *Project Manager* and be S1 continuously rated for use both when submerged in sewage at ambient temperatures and when in air.

Comply

(iv) The stator windings and stator leads shall be insulated with moisture resistant Class H insulation rated for 180 degree C (355 degree F) unless otherwise accepted by the *Project Manager*.

Comply

(v) Motor and pump shall be designed and produced by the same manufacturer unless otherwise accepted by the *Project Manager*.

Comply with surface cooling, cooling jacket is not necessary for submersible operation

(vi) Motors shall be provided with:

- built-in independent oil cooling jacket; or

- cooling jacket and cooled by pumped liquid

as the motor cooling system unless otherwise permitted by the *Project Manager*.

Comply (iv) Two mechanical seals shall be fitted on the shaft between the electric motor and the pump impeller. The space between these two seals shall be oil filled and an oil seepage detector shall be provided.

Comply (v) Motor terminals shall be contained in a separate IP68 protection terminal box for up to 20m submergence unless otherwise accepted by the *Project Manager*.

Comply (vi) Positive temperature coefficient thermistor or bimetal switch 2x to BS EN 60034-11 shall be embedded in the stator winding, together with the corresponding control relay, it shall protect the motor from being overheated. The control relay shall be recommended by the pump manufacturer and to be installed at the switchboard.

Comply (vii) Motor feed cables shall be ethylene propylene rubber insulated flexible cable, with each power core of tinned copper wire nylon earth screened and polychloroprene sheathed overall.

By other (viii) Emergency push buttons shall be installed in the vicinity of the motor.

By other (ix) The control circuit of motor shall incorporate the following protection features: -

- Time delay undervoltage protection
- Overload protection
- Lock rotor protection
- Phase failure protection
- Emergency push-button
- Motor overheat protection

By other (x) The *Contractor* shall be responsible to maintain the overall power factor for the electrical equipment under this contract to be not less than 0.85 lagging. This provision shall include but not limited to the supply and installation of power factor correction capacitors, controlling relay and contactors, etc.

(4) Accessories

By other (a) Pressure transducers for pump discharge pressures shall be provided for connection to the SCADA system.

By other (b) All the cables of the pump accessories shall be terminated at a water tight connection box mounted on the pump body.

By other (c) Each pump shall be equipped with a local control and instrumentation panel at pump level or nearest dry level if the pump is installed in wet

well. Each local control and instrumentation panel shall house the following instruments:

- Discharge pressure gauges.

By other (5) Control of Pumps

(a) General

(i) Each pump shall have the following modes of operation:

- SCADA- Auto;
- SCADA – Manual;
- Manual – Local; and
- Manual – Field.

The SCADA mode shall be selected by the "SCADA/ Off/Manual" key selector on the switchboard. The "SCADA-Auto" and "SCADA-Manual" shall be selected on the operator interface of the SCADA system with the SCADA mode selected. The "Manual - Local" and "Manual-Field" shall be selected by the "Local/ Field" selector at the switchboard with "SCADA/Off /Manual" at the "Manual" Position

(ii) The control status of each pump shall be sent to the SCADA system for control and indication purposes.

(iii) On pressing the emergency stop adjacent to a pump or drive motor, the drive motor shall stop and removed from the duty list immediately regardless of its mode of operation.

(iv) When the low level alarm is activated, all pumps shall be rendered inoperative irrespective of its mode of operation. A low level alarm shall be activated on the main control panel and the SCADA system.

(v) An adjustable time delay device shall be provided for each pump so that it cannot start immediately after stopping.

(vi) The pump shall be tripped and removed from the duty list, irrespective of mode of operation, when: -

- No flow in the discharge pipework is detected;
- Moisture is detected in the oil chamber of the mechanical seal;
- Thermal overload;
- Any abnormalities detected in the motor controller and the SCADA system;
- Excessive bearing temperature; and
- Excessive vibration.

(b) SCADA – Auto Control

(i) The SCADA system shall be able to control the operation of the equipment via the operator interface.

(ii) The pump shall be rendered inoperative when its corresponding

inlet and discharge gate valves are closed. When the actuator for the gate valve is power off or fail to communicate to the SCADA system, the SCADA system shall treat it as at 'closed' position.

(iii) When a stormwater pump is not installed or out of services due to maintenance or it cannot start due to faults of the pumpset or closure of valves, fault message shall be displayed on the operator interface and that dry weather flow pump shall be removed from the duty list.

(iv) A staggered starting feature shall be provided to prevent more than one duty pump from starting simultaneously in any event, in particular, on restoration of power supply after a mains failure. Adjustable timers shall be provided to accomplish this protection.

(v) The automatic control system shall initiate the cut-in and cut-out of the duty pumps in accordance with the preset water level as specified in the relevant clauses of this Particular Specification. The control system shall be designed so that the pumps shall work on rotational basis as the first duty, second duty, etc. The pump which cuts in first shall cut out first. The duty and standby pumps shall be started and stopped when the specified levels are reached for an adjustable time (0 to 1 minute).

(vi) The SCADA system shall allow the selection of duty pumps and the removal of pumps from the duty list.

(vii) When the water level persists above a specified level for an adjustable time (0 to 5 minutes), the SCADA system shall send an alarm signal on the screen of the operator interface and request for starting the standby pump.

(viii) When the water level is below a specified level for an adjustable time (0 to 5 minutes), the SCADA system shall stop all pumps from operation and send an alarm signal on the screen of the operator interface.

(ix) When the water level is above a specified level for an adjustable time (0 to 5 minutes), the SCADA system shall send a 'high level' alarm signal on the screen of the operator interface and close the inlet/overflow penstocks.

(x) The SCADA system shall get the flow information from the electromagnetic flowmeter.

(c) SCADA – Manual Control

(i) The mode of operation shall be selected via the operator interface of the SCADA system with "SCADA/Off/Manual" at the "SCADA" position at the switchboard.

(ii) The pumps shall be operated and stopped via the operator interface of the SCADA system.

(d) Manual-Local and Manual-Field

(i) The mode of operation shall be selected by the "Local/ Field" selector with the "SCADA/Off/Manual" at the "Manual" position on the main control panel. "Manual-Local" mode of operation means operation actuated by the „start" „stop" pushbuttons on the main control panel. "Manual-Field" mode of operation means operation actuated by the "start" "stop" pushbuttons on the local push button panel.

(6) Factory Test

(a) General

Comply (i) All stormwater pumps shall be subjected to inspections and tests during and after manufacture and before dispatch from the manufacturer's works, including inspection and testing of materials and castings, inspection of components and workmanship during manufacture, performance testing after manufacture and inspection of paintwork and packing.

NA, duck foot bend instead (ii) Pump shall be fitted with the suction bend supplied under the Contract or otherwise permitted by the *Project Manager*.

Comply (iii) The *Contractor* shall submit a testing plan at least three (3) months in advance before the test is carried out for the *Project Manager'*s consideration.

Comply (iv) The *Contractor* shall submit test reports for the *Project Manager*'s acceptance prior to pumps delivery.

Comply (v) The acceptance of the results of performance tests of the pumpsets by the *Project Manager* or his representative shall in no way be constructed as being final acceptance or as relieving of responsibility for the pumpsets.

Comply (vi) All the cost for conducting the tests mentioned shall be included in the Contract sum.

By other (vii) Commissioning shall be carried out in accordance with the relevant Clauses of this Particular Specification.

Comply (viii) The *Contractor* shall appoint an independent inspection body (IIB) to carry out the factory test of the pumps.

(b) Material Tests

Test reports of the following tests shall be submitted to the *Project Manager* or his representative within two weeks after the test has been completed:

Comply (i) Chemical analyses of casting of pump casing, suction cover, impeller, stuffing box cover and bearing pedestal to verify that the compositions comply with those of approved materials for pump construction.

Comply (ii) Test certificates for pump shaft and wear ring materials.

(c) Hydrostatic Pressure Tests

Hydrostatic pressure test on each pump body shall be performed in the presence of the *Project Manager*. The test shall be as follows:

Comply (i) Test pressure shall be the greater of the following:

- 150% of the maximum pressure which will occur in the pump body when the pump is operated at rated speed with the pump discharge valve fully closed and maximum suction head.

Comply (ii) The pump body shall be blanked off at the suction and discharge flanges and at the stuffing box cover where the gland is located, and shall not be restrained during the test.

Comply (iii) The test pressure shall be maintained for at least 5 minutes during which time there shall be no leakage from all joints and seals.

(d) Performance Tests

Comply (i) The test shall be performed prior to the application of final coating and in accordance BS EN ISO 9906 using clean cold water. The following curves for each pump at speeds to be selected by the *Project Manager* shall be developed from the test:

- Pump Total Head vs Flow;
- Pump Output Power vs Flow;
- Pump Net Positive Suction Head (NPSH) vs Flow;
- Motor Input Power vs Flow; and
- Overall Efficiency vs Flow.

Comply (ii) The test shall be carried out rated speed setting, one for the maximum and one for the minimum of the operating range. Test at each speed shall use not less than five test points to establish the above curves. These five points shall cover the entire flow range from close to zero discharge to Qmax or to the maximum value permitted by the test motor. At least four of the five points shall cover the required pumping range of the pump, Five measurement points are evenly distributed along the operating range of flowrate.

Comply (iii) All the operating points stated in the Performance Schedule shall be obtained during the factory test.

Comply (iv) Test configuration, procedures and certificates of instruments shall be submitted for acceptance by the *Project Manager* two months before commencement of the test.

Comply, noise measurement is not applicable for submersible operation (v) Noise and vibration measurements shall be carried out on each pumpset during the performance test. Noise measurements shall

NA for noise measurement

comply with ISO 9614-2 while vibration measurements shall comply with ISO 9908.

Comply (vi) The performance shall demonstrate the followings:

- Dynamic stability of the pump operation over the range of flow and speed;
- Compliance with the vibration severity limits; and
- Free from overheating / cavitations over the entire range of flow.

Comply (e) Factory testing – Non-compliance

Any pump will be liable for rejection by the *Project Manager* for any of the following reasons:

(i) Failure to pass hydrostatic pressure tests;

(ii) Failure to pass cavitation tests;

(iii) Failure to pass noise and vibration tests; and

(iv) Failure to meet the tolerance limits imposed on the guaranteed pump head and capacity as specified BS EN ISO 9906.

(7) Pre-commissioning Test

By other (a) Following completion of installation of the pumpsets and associated equipment, each pump shall undergo pre-commissioning tests using sewage/clean water to demonstrate proper operation. All ancillary equipment and associated instrumentation shall be checked to ensure correct functioning. Pre-commissioning tests shall include preliminary tests and proving tests.

(b) Prior to preliminary testing the *Contractor* shall satisfy the *Project Manager* that all pumps are installed properly, pipework and transmission assemblies are aligned correctly.

(c) Preliminary tests shall include cleaning, flushing and hydrostatic pressure testing of all associated small bore pipework, including bearing cooling water pipework and stuffing box flushing water pipework. Hydrostatic test pressure shall be 1.5 times of the maximum working pressures in the components to be tested.

(d) Proving tests shall be carried out on each pumpset and shall include the following:

(i) Performance tests to establish installed pump performance data over the entire “Pump Total Head vs Flow” curve, and to demonstrate dynamic stability of the pump operation over the range of flow;

(ii) Vibration monitoring to demonstrate compliance with approved vibration severity limits;

(iii) Reverse rotation of the pump due to backflow;

(iv) Simulation of power failure and emergency; and

(v) 24 hours, or a period less than 24 hours agreed by the *Project Manager*, endurance test at duty point.

(e) Performance proving tests shall be carried out on one pumpset at a time by pumping sewage/clean water by recirculation or proposed by the *Contractor* and acceptance by the *Project Manager*. Performance data to be obtained shall include those stated in PS Clauses 41.02 to 41.05 except for the NPSH and torque test. The purpose of the performance proving tests is to confirm the factory test results and verify the performance of the pumpsets under site conditions over the full operating range. In the event that the proving test results are not satisfactory compared to the factory test results, the *Contractor* shall immediately rectify the discrepancy.

(f) The *Contractor* shall demonstrate that the total head losses within the suction and discharge pipework, including valves and fittings, are within the values calculated by the *Contractor*. In the event that the measured head losses are higher than the calculated values, the *Contractor* shall rectify the discrepancy.

Sump Pump 41.57 (1) General

Comply (a) The pumps shall have stable, non-overloading and non-clogging characteristics. The pump shall be suitable for continuous running with great variations in water/sewage level.

Comply (b) The pumps shall be suitable for pumping raw sewage containing rags, fibrous material and general waste matters including grit and other abrasives.

Comply (c) Each component casting of the pump body shall have extra wall thickness to compensate for the corrosive and abrasive action of the sewage over 25 years operation in addition to the compliance with the strength requirement.

Comply (d) The pumps for Argyle Street Playground Stormwater Storage Scheme (SUP-201 and SUP-202) are required to deliver not less than 15 $L/_S$ against 27m head when operating in 1 1 Duty + 1 Standby mode.

Comply (e) The pump for Packaged Pump Scheme at Dock Street, Hung Hom (SUP-801) are required to deliver not less than 11 $L/_S$ against 6m head. The pump shall be portable type.

Comply,
double guide rails by other

(f) For the wet well installation, guide rail systems shall be completed with double guide rails, guide holders, lifting chain, chain bucket and lifting shackle and all accessories necessary for lifting the pump up to floor level. For the wet well installation, lifting chains shall include shackles for connection to pump lifting points. Lifting chains shall be at least 1.5m longer than the guide rails. Additional stainless steel rings shall be attached to the chain in an interval of 2m to facilitate the lifting

of equipment by lifting appliances.

Comply (g) The pumps shall be compatible with 380V 3 phases 50Hz power supply system.

Comply (h) The pump motor shall be induction type with a squirrel cage rotor, housed in an air filled watertight chamber submersible up to 20m according IEC 60034 protection class IP 68. The stator windings and stator leads shall be insulated with moisture resistant Class H insulation rated for 180 degree C (355 degree F). Motor and pump shall be designed and produced by the same manufacturer. The motor shall be S1 continuously rated for use both when submerged in sewage at ambient temperatures and when in air.

Comply (i) The pumps shall be single-stage close-coupled submersible centrifugal type with of semi open multi vane impeller designed to transport wastewater with fibrous materials and heavy sludge unless otherwise approved by the *Project Manager*. The impeller blades shall be self-cleaning upon each rotation as they pass across a sharp relief groove in the Insert ring and shall keep the impeller blades clear of debris.

Comply (j) Casing shall have sufficient metal thickness in the volute where not protected by wearing plates to compensate for corrosive and abrasive action of the sewage.

(k) Material of construction shall be as follows:

Comply (i) Pump casing: Stainless Steel grade 316 to BS EN 10088, Cast Iron Grade EN-GLJ-250 or accepted better material;

Comply (ii) Impeller: Duplex Stainless Steel ASTM CD-4MCuN or accepted better material;

Comply (iii) Motor casing: Stainless Steel grade 316 to BS EN 10088, Cast Iron Grade EN-GLJ-250 or accepted better material;

Comply (iv) Shaft: Stainless Steel, grade 316 to BS EN 10088, Stainless Steel ASTM-AISI 431 or accepted better material;

Comply (v) Shaft sleeve (if applicable): Stainless Steel grade 316 to BS EN 10088, Cast Iron Grade EN-GLJ-250 or accepted better material;

Comply (vi) Seals: Double mechanical seal;

Comply (vii) Bolts and nuts: Stainless Steel grade 316 to BS EN 10088, Cast Iron Grade EN-GLJ-250 or accepted better material; and

Comply (viii) Bearings: Maintenance free ball bearings and permanently lubrication by grease

Comply (g) Lifting eye bolts shall be fitted to casing for connection of lifting chains. Lifting chains shall be of stainless steel and have shackles for connection to the pump lifting eye bolts and the hook of a hoist.

By other (2) Control of Pumps

(a) Each pump shall have the following modes of operation: